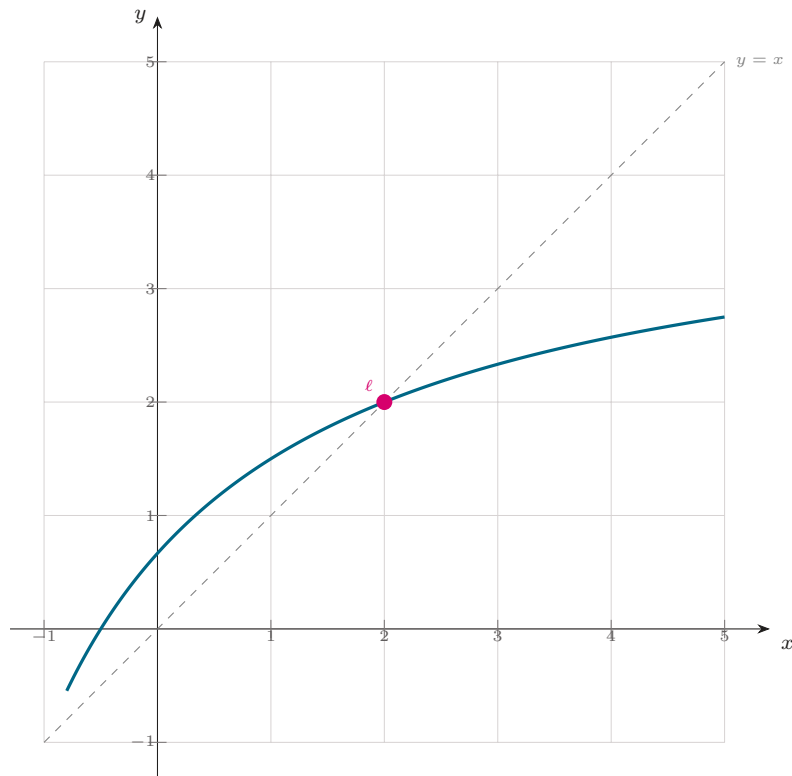


Corrigé — Exercice 18

A. 1) $f'(x) = \frac{4(x+3) - (4x+2)}{(x+3)^2} = \frac{10}{(x+3)^2} > 0$: croissante. **2)** $f(2) = \frac{10}{5} = 2$; $f([0, 2]) = [f(0), f(2)] = [\frac{2}{3}, 2] \subset [0, 2]$. **B. 3)** $u_0 = 0 \in [0, 2]$; si $0 \leq u_n < 2$ alors $u_{n+1} = f(u_n) \in [\frac{2}{3}, 2]$. **4)** $v_{n+1} = \frac{u_{n+1} - 2}{u_{n+1} + 1} = \frac{2u_n - 4}{5u_n + 5} = \frac{2}{5}v_n$: géométrique de raison $\frac{2}{5}$, $v_0 = -2$. **5)** $v_n = -2 \left(\frac{2}{5}\right)^n$; $u_n = \frac{2 + v_n}{1 - v_n} = \frac{2 - 2\left(\frac{2}{5}\right)^n}{1 + 2\left(\frac{2}{5}\right)^n}$. **6)** $\left(\frac{2}{5}\right)^n \rightarrow 0$ donc $\lim u_n = 2$.



$C_f : y = \frac{4x+2}{x+3}$, $y = x$ et le point fixe $\ell = 2$ (Exercice 18).